



**IUS**  
INSTITUT  
UNIVERSITAIRE  
DES SCIENCES

**INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)**

**Faculté des sciences et des technologies (FST)**

**Rapport laboratoire TD N° 8– Réseaux I**

**Préparé par : Brian ANTOINE**

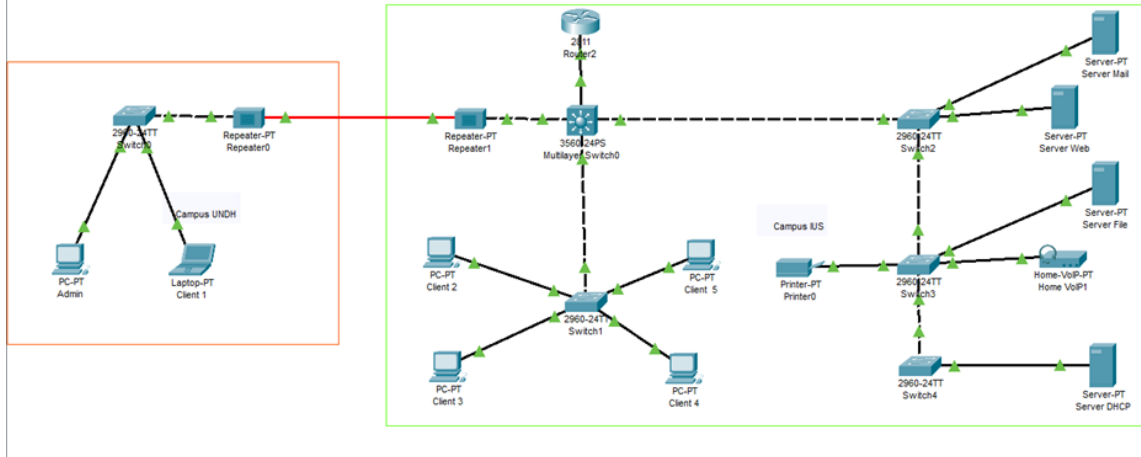
**Présenté à: M. Ismaël SAINT AMOUR**

**Niveau : L3**

**Date: 04/01/2026**

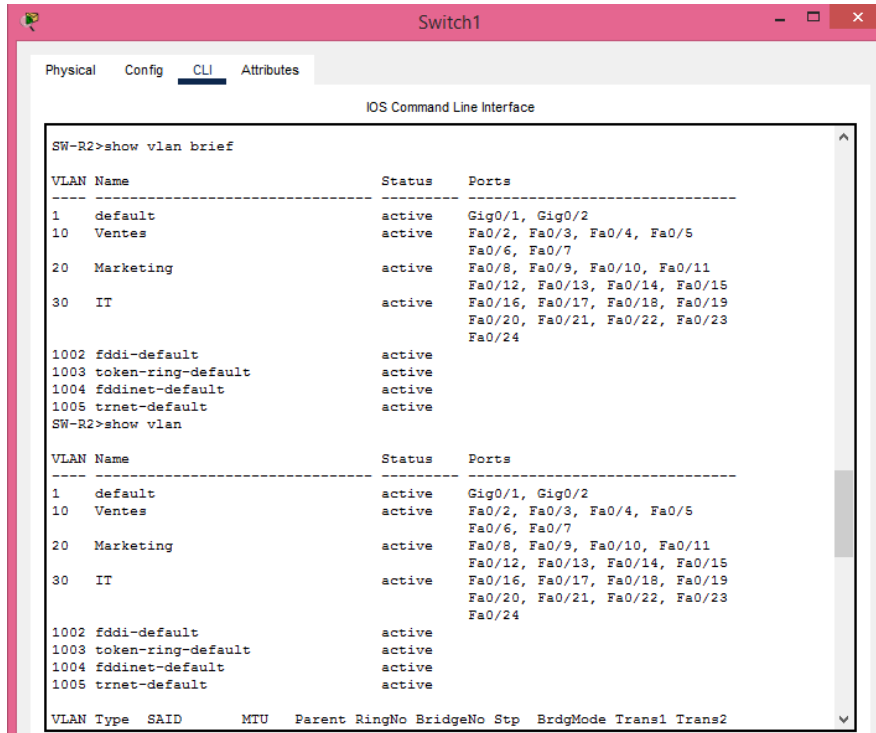
1)

❖ Reproduction de la topologie :



## ➤ Tests de vérifications

### ■ Vérification des Vlan :



The screenshot shows a network switch interface titled "Switch1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The command "SW-R2>show vlan brief" has been entered, resulting in a table of VLAN configurations. Below this, the command "SW-R2>show vlan" is entered, followed by another table showing detailed VLAN information. At the bottom, a header for a more detailed table is visible.

```
SW-R2>show vlan brief
```

| VLAN | Name               | Status | Ports                          |
|------|--------------------|--------|--------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                 |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5     |
|      |                    |        | Fa0/6, Fa0/7                   |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11   |
|      |                    |        | Fa0/12, Fa0/13, Fa0/14, Fa0/15 |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19 |
|      |                    |        | Fa0/20, Fa0/21, Fa0/22, Fa0/23 |
|      |                    |        | Fa0/24                         |
| 1002 | fddi-default       | active |                                |
| 1003 | token-ring-default | active |                                |
| 1004 | fddinet-default    | active |                                |
| 1005 | trnet-default      | active |                                |

```
SW-R2>show vlan
```

| VLAN | Name               | Status | Ports                          |
|------|--------------------|--------|--------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                 |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5     |
|      |                    |        | Fa0/6, Fa0/7                   |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11   |
|      |                    |        | Fa0/12, Fa0/13, Fa0/14, Fa0/15 |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19 |
|      |                    |        | Fa0/20, Fa0/21, Fa0/22, Fa0/23 |
|      |                    |        | Fa0/24                         |
| 1002 | fddi-default       | active |                                |
| 1003 | token-ring-default | active |                                |
| 1004 | fddinet-default    | active |                                |
| 1005 | trnet-default      | active |                                |

| VLAN Type | SAID | MTU | Parent | RingNo | BridgeNo | Stp | BrdgMode | Trans1 | Trans2 |
|-----------|------|-----|--------|--------|----------|-----|----------|--------|--------|
|-----------|------|-----|--------|--------|----------|-----|----------|--------|--------|

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```

SW-R2>show vlan

```

| VLAN | Name               | Status | Ports                                                                      |
|------|--------------------|--------|----------------------------------------------------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                                                             |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5<br>Fa0/6, Fa0/7                                 |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15             |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24 |
| 1002 | fddi-default       | active |                                                                            |
| 1003 | token-ring-default | active |                                                                            |
| 1004 | fddinet-default    | active |                                                                            |
| 1005 | trnet-default      | active |                                                                            |

VLAN Name Status Ports

|      |                    |        |                                                                            |
|------|--------------------|--------|----------------------------------------------------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                                                             |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5<br>Fa0/6, Fa0/7                                 |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15             |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24 |
| 1002 | fddi-default       | active |                                                                            |
| 1003 | token-ring-default | active |                                                                            |
| 1004 | fddinet-default    | active |                                                                            |
| 1005 | trnet-default      | active |                                                                            |

VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2

|   |      |        |      |   |   |   |   |   |   |   |
|---|------|--------|------|---|---|---|---|---|---|---|
| 1 | enet | 100001 | 1500 | - | - | - | - | - | 0 | 0 |
|---|------|--------|------|---|---|---|---|---|---|---|

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```

SW-R2>show vlan brief

```

| VLAN | Name               | Status | Ports                                                                      |
|------|--------------------|--------|----------------------------------------------------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                                                             |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5<br>Fa0/6, Fa0/7                                 |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15             |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24 |
| 1002 | fddi-default       | active |                                                                            |
| 1003 | token-ring-default | active |                                                                            |
| 1004 | fddinet-default    | active |                                                                            |
| 1005 | trnet-default      | active |                                                                            |

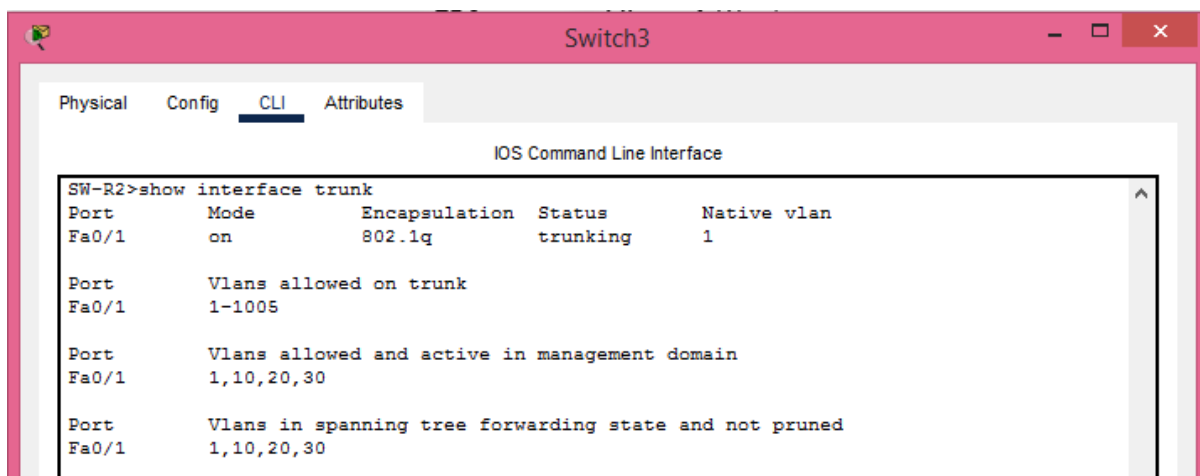
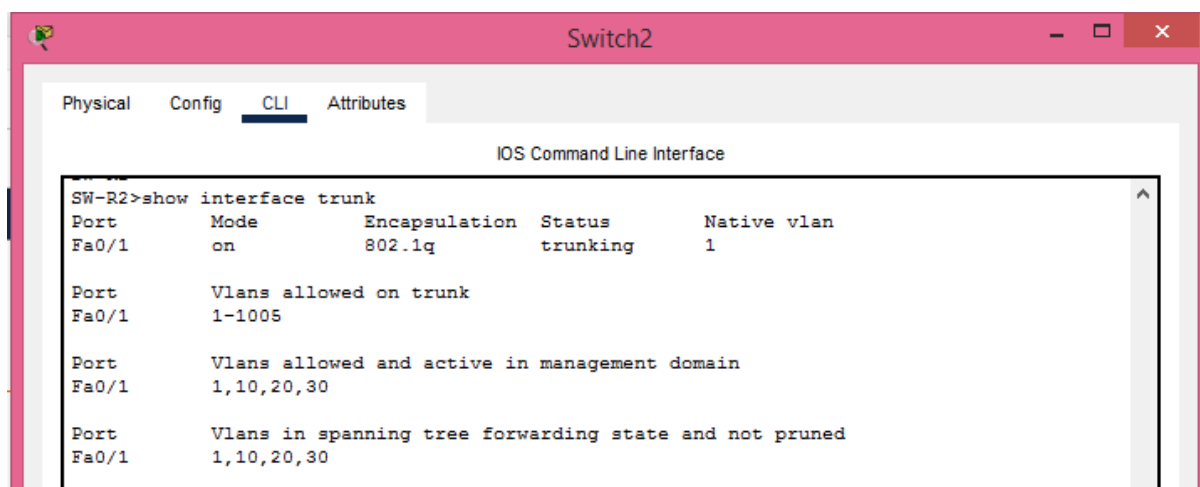
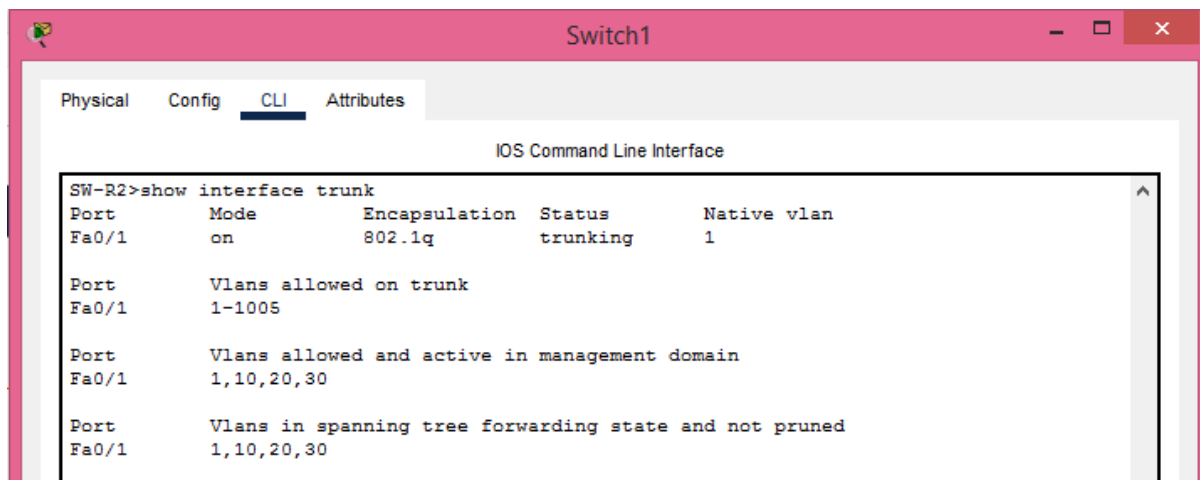
VLAN Name Status Ports

|      |                    |        |                                                                            |
|------|--------------------|--------|----------------------------------------------------------------------------|
| 1    | default            | active | Gig0/1, Gig0/2                                                             |
| 10   | Ventes             | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5<br>Fa0/6, Fa0/7                                 |
| 20   | Marketing          | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15             |
| 30   | IT                 | active | Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24 |
| 1002 | fddi-default       | active |                                                                            |
| 1003 | token-ring-default | active |                                                                            |
| 1004 | fddinet-default    | active |                                                                            |
| 1005 | trnet-default      | active |                                                                            |

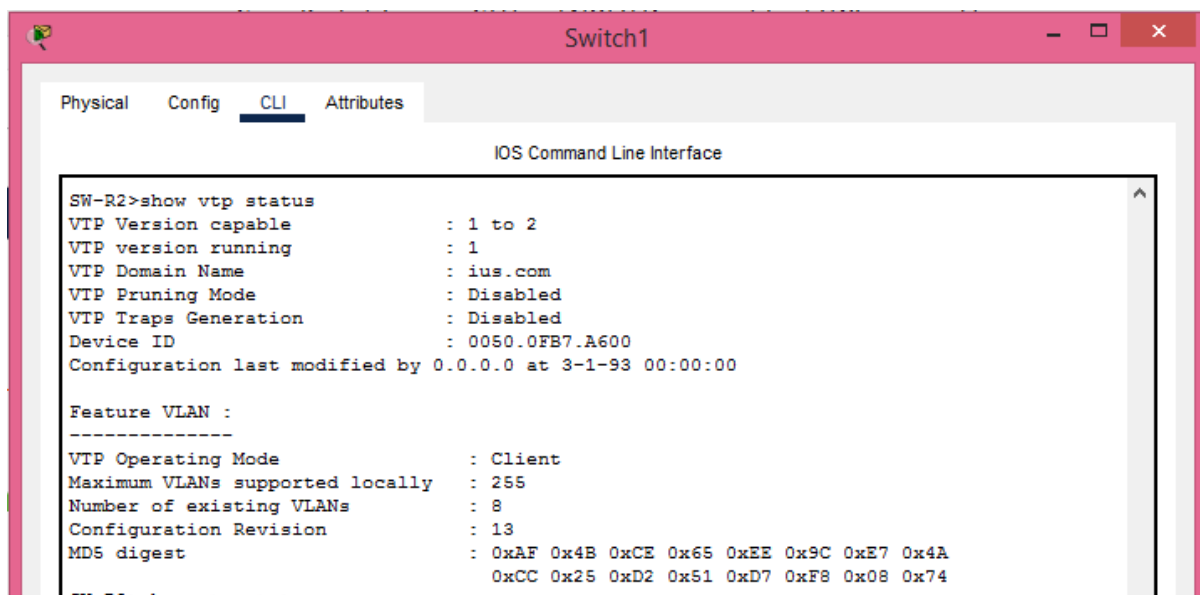
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2

|   |      |        |      |   |   |   |   |   |   |   |
|---|------|--------|------|---|---|---|---|---|---|---|
| 1 | enet | 100001 | 1500 | - | - | - | - | - | 0 | 0 |
|---|------|--------|------|---|---|---|---|---|---|---|

- Vérification des trunks :



- Vérification VTP :



The screenshot shows a network switch configuration window titled "Switch1". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The command "SW-R2>show vtp status" has been entered, resulting in the following output:

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0050.0FB7.A600
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision   : 13
MD5 digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                          : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.C70D.E700
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision   : 13
MDS digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                        : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0002.4ABC.3200
Configuration last modified by 0.0.0.0 at 3-1-93 00:13:29

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision   : 0
MDS digest               : 0x6B 0x43 0xE6 0x23 0xB2 0xC6 0xD2 0x0B
                        : 0x5C 0x0F 0x11 0x3A 0xE7 0xF3 0x5D 0xE9
```

- Vérification STP :

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```

4 vlans                0      0      0      8      8

SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

| Name     | Blocking | Listening | Learning | Forwarding | STP Active |
|----------|----------|-----------|----------|------------|------------|
| VLAN0001 | 0        | 0         | 0        | 1          | 1          |
| VLAN0010 | 0        | 0         | 0        | 1          | 1          |
| VLAN0020 | 0        | 0         | 0        | 5          | 5          |
| VLAN0030 | 0        | 0         | 0        | 1          | 1          |
| 4 vlans  | 0        | 0         | 0        | 8          | 8          |

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```

SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

```

| Name     | Blocking | Listening | Learning | Forwarding | STP Active |
|----------|----------|-----------|----------|------------|------------|
| VLAN0001 | 0        | 0         | 0        | 1          | 1          |
| VLAN0010 | 0        | 0         | 0        | 2          | 2          |
| VLAN0020 | 0        | 0         | 0        | 1          | 1          |
| VLAN0030 | 0        | 0         | 0        | 3          | 3          |
| 4 vlans  | 0        | 0         | 0        | 7          | 7          |



Switch3

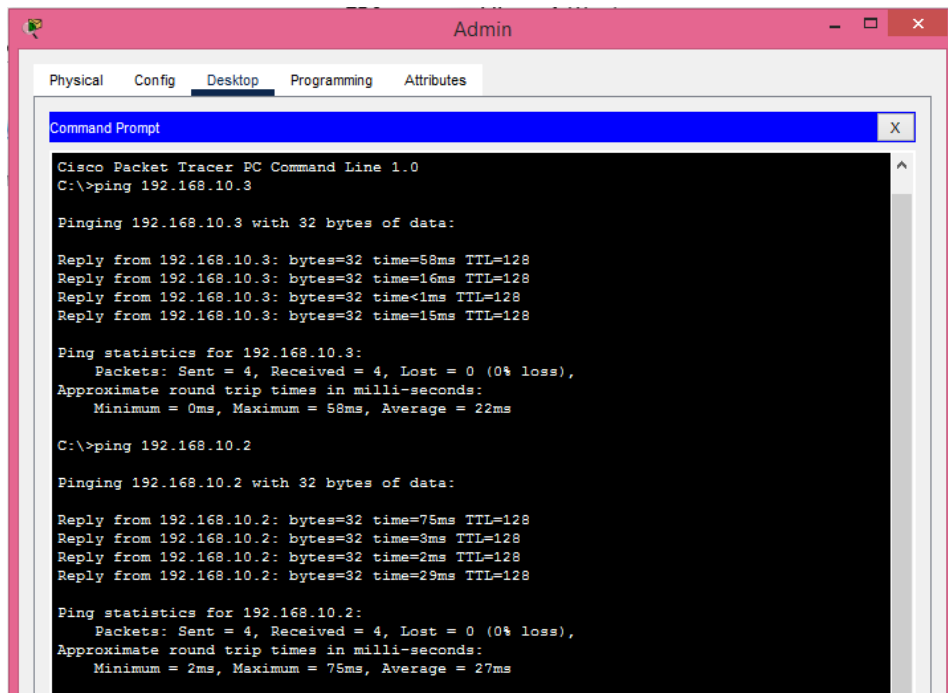
Physical Config CLI Attributes

IOS Command Line Interface

SW-R2>show spanning-tree summary  
Switch is in pvst mode  
Root bridge for: Marketing IT  
Extended system ID is enabled  
Portfast Default is disabled  
PortFast BPDU Guard Default is disabled  
Portfast BPDU Filter Default is disabled  
Loopguard Default is disabled  
EtherChannel misconfig guard is disabled  
UplinkFast is disabled  
BackboneFast is disabled  
Configured Pathcost method used is short

| Name     | Blocking | Listening | Learning | Forwarding | STP Active |
|----------|----------|-----------|----------|------------|------------|
| VLAN0001 | 0        | 0         | 0        | 1          | 1          |
| VLAN0010 | 0        | 0         | 0        | 2          | 2          |
| VLAN0020 | 0        | 0         | 0        | 1          | 1          |
| VLAN0030 | 0        | 0         | 0        | 4          | 4          |
| 4 vlans  | 0        | 0         | 0        | 8          | 8          |

- ❖ Testons la connectivité
  - ✓ Test de connectivite sur Vlan 10 :



The screenshot shows the Cisco Packet Tracer Admin window with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of two ping commands. The first command is 'ping 192.168.10.3', which shows four successful replies with varying times (58ms, 16ms, <1ms, 15ms) and a TTL of 128. The statistics for 192.168.10.3 show 4 packets sent, 4 received, 0% loss, and an average round trip time of 22ms. The second command is 'ping 192.168.10.2', which also shows four successful replies with times (75ms, 3ms, 2ms, 29ms) and a TTL of 128. The statistics for 192.168.10.2 show 4 packets sent, 4 received, 0% loss, and an average round trip time of 27ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=58ms TTL=128
Reply from 192.168.10.3: bytes=32 time=16ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 22ms

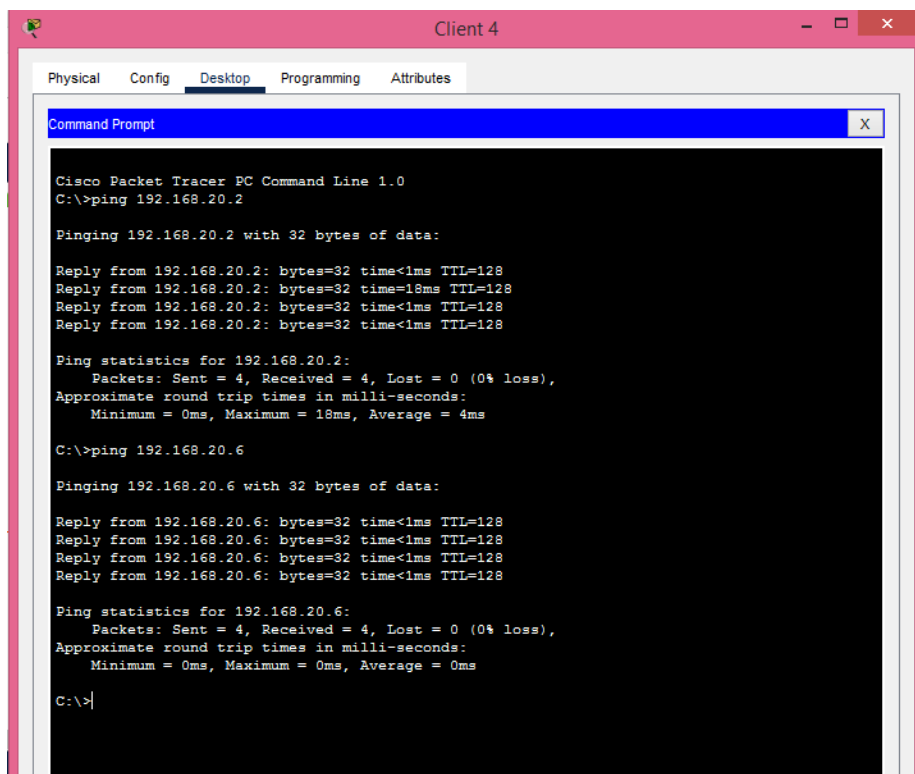
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=75ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time=29ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 75ms, Average = 27ms
```

✓ Test de connectivite sur Vlan 20 :



The screenshot shows the Cisco Packet Tracer Client 4 window with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of two ping commands. The first command is 'ping 192.168.20.2', which shows four successful replies with times (<1ms, 18ms, <1ms, <1ms) and a TTL of 128. The statistics for 192.168.20.2 show 4 packets sent, 4 received, 0% loss, and an average round trip time of 4ms. The second command is 'ping 192.168.20.6', which also shows four successful replies with times (<1ms, <1ms, <1ms, <1ms) and a TTL of 128. The statistics for 192.168.20.6 show 4 packets sent, 4 received, 0% loss, and an average round trip time of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=18ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>ping 192.168.20.6

Pinging 192.168.20.6 with 32 bytes of data:

Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

✓ Test de connectivite sur Vlan 30 :

